AI Test Exam

1. **Question 1:**

Consider this sequence of model accuracies during training:

Epoch	1	2	3	4	5	6	7	8	9
Train Acc	0.6	0.64	0.7	0.76	0.85	0.88	0.9	0.93	0.96
Val Acc	0.52	0.59	0.66	0.7	0.76	0.83	0.79	0.74	0.7

a) What is the most likely problem occurring in this model?
b) Name two techniques you would apply to reduce this problem in this scenario.
c) What is the best epoch to select the model for deployment? Why?
d) If validation accuracy kept increasing together with training accuracy, what would that indicate about the model?

2. **Question 2:**

You are solving a binary classification problem:

- 90% negative
- 10% positive

a) Describe two precise training techniques to handle imbalance.
b) Explain mathematically or intuitively why it works.
c) Would you apply it at test time? Why or why not?

3. **Question 3:**

You are building an AI system to detect fraudulent invoices. Missing fraud is very costly (each missed fraud costs $50,000), while false alarms cost $500 to investigate. Which of the following is the most appropriate evaluation metric?

- Accuracy
- Precision
- Recall
- Loss value

Explain why

-

4. **Question 4:**

What is the purpose of using 1×1 convolution?

The answer must include:

- The primary computational benefit
- An example use case in a popular architecture
- How it affects model capacity vs. computation trade-off

5. **Question 5:**

Explain the difference between:

- BERT-style embedding models
- Generative LLM embeddings

The answer must include:

- Architecture: How the neural network is structurally designed to read text.
- Model size: Number of parameters → determines memory, speed, and cost.
- Training Objective: Training Method and what task the model was trained to optimize.

6. **Question 6:**

Your team deploys a Vietnamese customer-support LLM. After launch, you observe:

- Answers are fluent and polite
- But sometimes contain subtle wrong facts (hallucinations)
- Users rarely report errors immediately
- You cannot retrain the base model for now

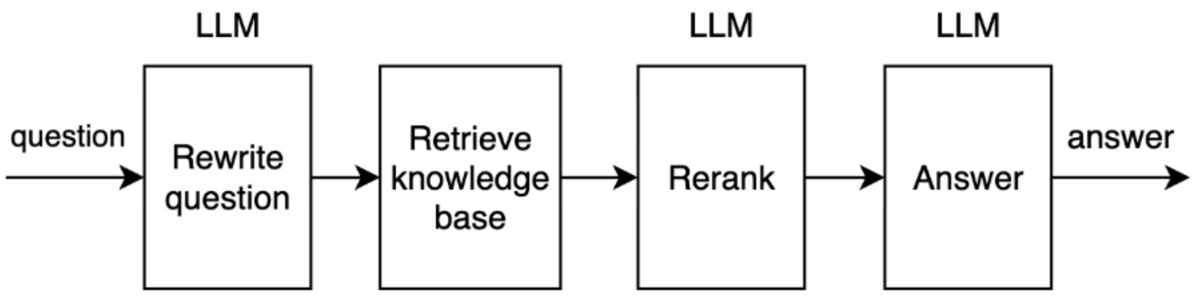
Propose a practical solution to reduce hallucinations in production under these constraints.

Your answer should include:

1. Your high-level idea.
2. How your method detects or prevents hallucinations.
3. One limitation or risk of your approach.
4. One metric you would track to verify improvement.

7. **Question 7:**

A flower shop needs a chatbot to help customers find information about its products. The following RAG pipeline is proposed:



The user's message is solely passed to a LLM to be rewritten to make it clearer. Then, the rewritten message is used to retrieve knowledge chunks from a knowledge base. A LLM is used to re-rank each chunk in sequence. The rewritten message and all reranked chunks are passed to a LLM to get the final answer.

What issues are there in this pipeline and how to improve it?

8. **Question 8:**

You are working on a Retrieval-Augmented Generation (RAG) system where the knowledge base consists of Markdown documents. There are about 300 documents and none of them has titles, headings, and subheadings in the content. The system currently uses semantic search to retrieve relevant chunks.

You observe the following issues:

- When a document contains a **long list or table** (e.g., a policy with 20 bullet points), and the user asks a broad question like *"Show me policy ABC"*, the system only returns a **partial answer** (e.g., only the first 4–5 bullet points).
- Content from **code blocks** is not well preserved after chunking, leading to missing or distorted information in the final answer.
- When users ask about information located near the end of a document, the system often fails to retrieve the relevant chunks, even though the content exists.

How do you solve those issues? Explain your solution.

9. **Question 9:**

Analyze the advantages and limitations (especially the issue of overfitting) of the traditional Decision Tree algorithm. Based on that, explain how Random Forest (Bagging method) and Boosting algorithms (such as AdaBoost and Gradient Boosting) address these limitations to create a more powerful predictive model.

10. **Question 10:**

For sequential data, why is the architecture of Recurrent Neural Networks (RNNs) suitable? However, basic RNNs suffer from a serious issue called the "vanishing gradient" problem. Explain the cause of this phenomenon and how the Long Short-Term Memory (LSTM) architecture addresses it.